Lecture plan · for the lecturer

Public-key cryptography

Cyber security · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Run Diffie–Hellman.
- Explain RSA encryption and signatures.
- Expose the man in the middle.

Activity: Alice, Bob and Eve

1. Warm-up (5 min): Ask students how a browser can share a secret with a server it has never met.
2. Mix (10 min): Show the paint-mixing analogy and why unmixing is hard.
3. Exchange (20 min): Groups of three run DH with p = 23; Eve records everything she hears.
4. Attack (20 min): Replace Eve with an active attacker, then fix it with signatures.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Why couldn't Eve find the key?
- What did signatures add?
- Which misconception did you hear today?

Exit ticket

- Say what encrypting to Bob's public key does and does not guarantee.

Try it in your next lecture
Put students in threes as Alice, Bob and Eve; they run Diffie–Hellman with p = 23 on paper while Eve tries to find the key.

Seminar sheet · for students

Public-key cryptography

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Name _____ Group _____ Date _____

1 **Plain Diffie–Hellman has no authentication. What can a network attacker do?**

- ◯ Nothing: the shared key never crosses the network
- ◯ Compute the key from A and B, which travel in the clear
- ◯ Run an exchange with each side and relay messages

2 **Run Diffie–Hellman with p = 23, g = 5.**

Alice picks a = 3, so A = 5^3 mod 23 = _____. Bob sends B = 4, so the key is 4^3 mod 23 = _____.

Word bank: 4 · 15 · 10 · 18

3 **Which statements are true? Choose all that apply.**

- ☐ TLS authenticates the key exchange with certificates
- ☐ Encrypting to Bob's public key proves Alice sent it
- ☐ A signature made with Alice's private key can be verified by anyone
- ☐ Textbook RSA without padding is insecure

4 **A man-in-the-middle attack on an unauthenticated exchange works whatever the length of the RSA key.**

◯ True ◯ False

5 **Match each goal to the operation that provides it.**

Only Bob can read it	• • • •	Sign with Alice's private key
Anyone can check Alice wrote it	• • • •	Encrypt with Bob's public key
Agree a key in public	• • • •	A certificate
Bind a public key to a name	• • • •	Diffie–Hellman

Answers · for the lecturer

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1 **Run an exchange with each side and relay messages**

Without authentication, an attacker can sit in the middle with two keys.

2 **Alice picks a = 3, so A = 5^3 mod 23 = 10. Bob sends B = 4, so the key is 4^3 mod 23 = 18.**

125 mod 23 = 10 and 64 mod 23 = 18. Bob's secret is b = 4 (5^4 mod 23 = 4), so he reaches the same key: 10^4 mod 23 = 18.

3 **A signature made with Alice's private key can be verified by anyone; Textbook RSA without padding is insecure; TLS authenticates the key exchange with certificates**

Anyone can encrypt to Bob's public key, so encryption alone proves nothing about the sender.

4 **True**

The flaw is missing authentication, not weak keys; the attacker never needs to break any key.

5 **Only Bob can read it — Encrypt with Bob's public key**
Anyone can check Alice wrote it — Sign with Alice's private key
Agree a key in public — Diffie–Hellman
Bind a public key to a name — A certificate

Asking what each operation guarantees targets the encryption-equals-authentication misconception.

For you to think about

How much number theory do your students need before they meet public-key cryptography? What could they do by hand first?